Classical Field Theory — Exercise Sheet 20

Nonabelian constraints and local reduction

Solutions

Use $T_a = -\mathrm{i}\sigma_a/2$ for $\mathfrak{su}(2)$, so $[T_a, T_b] = \epsilon_{abc}T_c$ and $\langle T_a, T_b\rangle = \delta_{ab}$. All spatial fields are smooth on $\mathbb{T}^3$. Write $\mathcal{V}_{\mathrm{per}} = (2\pi)^3$ for its volume. The canonical bracket has the order fixed in Lectures 7 and 20.

Exercise 20.1 (A homogeneous constraint manifold). Restrict the spatial fields to constants $A_i = a_i^a T_a$ and $\Pi_i = p_i^a T_a$. A parameter $\varepsilon = \varepsilon^a T_a$ is also constant.

(a) Pull the canonical symplectic form back to these constants and derive their finite-dimensional bracket. Compute the generator $C[\varepsilon] = \mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, \varepsilon]\rangle$ and its Hamiltonian vector field. Prove its constraint bracket, including the volume factor and sign.

(b) At $A_i = T_i$ for $i = 1, 2, 3$, write $\Pi_i = \sum_j M_{ij} T_j$ for a real matrix M. Classify all M satisfying $\sum_i [A_i, \Pi_i] = 0$. Give one nonscalar matrix in the constraint set and one outside it. Explain why checking a single homogeneous state cannot prove the constraint algebra on the full periodic field space.

Solution. The canonical form on the homogeneous subspace is

$$\Omega_{\mathrm{hom}} = \mathcal{V}_{\mathrm{per}} \sum_{i,a} \mathrm{d}a_i^a \wedge \mathrm{d}p_i^a.$$

Solving $\iota_{X_F}\Omega_{\mathrm{hom}} = \mathrm{d}F$ gives $X_F = (\mathcal{V}_{\mathrm{per}}^{-1} F_p, -\mathcal{V}_{\mathrm{per}}^{-1} F_a)$ and therefore

$$\{F, G\}_{\mathrm{hom}} = \frac{1}{\mathcal{V}_{\mathrm{per}}} \sum_{i,a} (F_{a_i^a} G_{p_i^a} - F_{p_i^a} G_{a_i^a}).$$

The reciprocal volume is essential.

Vary the displayed generator, holding its parameter fixed. Invariance of the Lie-algebra pairing gives

$$(C[\varepsilon])_{\Pi_i} = \mathcal{V}_{\mathrm{per}}[A_i, \varepsilon], \qquad (C[\varepsilon])_{A_i} = \mathcal{V}_{\mathrm{per}}[\varepsilon, \Pi_i].$$

The reciprocal volume in the bracket cancels these factors and gives the variations

$$\delta A_i = [A_i, \varepsilon], \qquad \delta \Pi_i = [\Pi_i, \varepsilon].$$

Substitution of both gradients into that bracket yields

$$\{C[\varepsilon], C[\zeta]\} = \mathcal{V}_{\mathrm{per}} \sum_i \big(\langle [\varepsilon, \Pi_i], [A_i, \zeta]\rangle - \langle [A_i, \varepsilon], [\zeta, \Pi_i]\rangle\big).$$

Move the outer parameter commutators with pairing invariance. The result is

$$-\mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [\varepsilon, [A_i, \zeta]] + [[A_i, \varepsilon], \zeta]\rangle = -\mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, [\varepsilon, \zeta]]\rangle.$$

The matrix Jacobi identity gives the last equality. This is $-C[[\varepsilon, \zeta]]$.

At the chosen configuration the Gauss array is

$$\sum_i [T_i, M_{ij}T_j] = (M_{23} - M_{32})T_1 + (M_{31} - M_{13})T_2 + (M_{12} - M_{21})T_3.$$

Its three coefficients vanish exactly when M is symmetric. For example,

$$M = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 0 & 3 \\ 0 & 3 & -1 \end{pmatrix}$$

satisfies the constraint and is not scalar. The matrix with $M_{12} = 1$ and every other entry zero instead gives T_3. These examples test a subset of phase space. The full field-space algebra also involves spatial derivatives of arbitrary smooth parameters and must be proved with their integration by parts.

Exercise 20.2 (Integrating a residual gauge generator). For a fixed smooth periodic $\varepsilon(x)$, let $h_s(x) = \exp(s\varepsilon(x))$ and define

$$A_i(s) = h_s^{-1}A_i h_s + h_s^{-1}\partial_i h_s, \qquad \Pi_i(s) = h_s^{-1}\Pi_i h_s.$$

Here s is the transformation parameter, not physical time.

(a) Prove that these arrays solve $\partial_s A_i(s) = D_i^{(s)}\varepsilon$ and $\partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon]$. Show that the transformation preserves the canonical form, the temporal-gauge Hamiltonian and the Gauss constraint surface.

(b) Take $\varepsilon = \sin x^1\, T_3$, $A_1 = T_1$, $A_2 = T_3$, $A_3 = 0$, $\Pi_1 = T_2$ and $\Pi_2 = \Pi_3 = 0$. Compute the transformed fields and their Gauss array. Is this example on the constraint surface? What changes if a finite gauge transformation depends on physical time?

Solution. The identities $\partial_s h_s = \varepsilon h_s$ and $\partial_s h_s^{-1} = -h_s^{-1}\varepsilon$ give

$$\partial_s A_i(s) = h_s^{-1}[A_i, \varepsilon]h_s + h_s^{-1}(\partial_i\varepsilon)h_s, \qquad \partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon].$$

Differentiating $h_s^{-1}\varepsilon h_s = \varepsilon$ in x^i gives

$$h_s^{-1}(\partial_i\varepsilon)h_s = \partial_i\varepsilon + [h_s^{-1}\partial_i h_s, \varepsilon].$$

Combining these two equations proves the asserted equation for $A_i(s)$.

For variations of the initial phase point the matrix h_s is fixed, so $\delta A_i(s) = h_s^{-1}\delta A_i h_s$ and $\delta\Pi_i(s) = h_s^{-1}\delta\Pi_i h_s$. Trace invariance preserves the canonical two-form. The product rule, with $\partial_i h_s^{-1} = -h_s^{-1}(\partial_i h_s)h_s^{-1}$, gives

$$\partial_i(h_s^{-1}Xh_s) = h_s^{-1}(\partial_i X)h_s - [h_s^{-1}\partial_i h_s, h_s^{-1}Xh_s].$$

Adding the connection commutator proves the covariant-derivative identity. For the curvature, put $b_i = h_s^{-1}\partial_i h_s$ and $\widehat{A}_i = h_s^{-1}A_i h_s$. The same inverse derivative gives

$$\partial_i\widehat{A}_j = h_s^{-1}(\partial_i A_j)h_s - [b_i, \widehat{A}_j], \qquad \partial_i b_j - \partial_j b_i = -[b_i, b_j].$$

The second equality follows by differentiating both b terms; their mixed second derivatives cancel. Since $A_i(s) = \widehat{A}_i + b_i$, substitution gives

$$\begin{aligned} F_{ij}(s) = h_s^{-1}(\partial_i A_j - \partial_j A_i)h_s - [b_i, \widehat{A}_j] + [b_j, \widehat{A}_i] - [b_i, b_j] \\ + [\widehat{A}_i + b_i, \widehat{A}_j + b_j] = h_s^{-1}F_{ij}h_s. \end{aligned}$$

Thus both the kinetic and curvature squares in the Hamiltonian are unchanged. The covariant-derivative identity gives $\mathcal{D}(s) = h_s^{-1}\mathcal{D}h_s$, preserving its zero set.

For the specified parameter, conjugation by $\exp(-\alpha T_3)$ rotates T_1 to $\cos\alpha T_1 - \sin\alpha T_2$ and T_2 to $\sin\alpha T_1 + \cos\alpha T_2$. The derivative of h_s contributes $s\cos x^1 T_3$ to $A_1(s)$. Thus the full nonzero arrays are

$$A_1(s) = \cos\alpha T_1 - \sin\alpha T_2 + s\cos x^1 T_3, \qquad A_2(s) = T_3,$$
$$\Pi_1(s) = \sin\alpha T_1 + \cos\alpha T_2.$$

The components $A_3(s)$, $\Pi_2(s)$ and $\Pi_3(s)$ are zero. Writing $\alpha' = s\cos x^1$, their derivative term is

$$\partial_1\Pi_1(s) = \alpha'(\cos\alpha\, T_1 - \sin\alpha\, T_2).$$

Its commutator with the derivative contribution to $A_1(s)$ is the negative of this expression. The remaining commutator of the two rotated T_1, T_2 directions is T_3. Hence $\mathcal{D}(s) = T_3$, also obtained by conjugating the original $\mathcal{D} = T_3$. Gauge invariance does not turn nonconstraint data into constraint data.

In temporal gauge $A_0 = 0$, a physical-time-dependent h gives $A_0' = h^{-1}\partial_t h$. It is zero exactly when $\partial_t h = 0$. The parameter flow above has no physical-time dependence.

Exercise 20.3 (A linearized reduced bracket and its limit). Fix a real constant η and the background $A_1^0 = \eta T_3$, $A_2^0 = A_3^0 = 0$, $\Pi^0 = 0$. Work only with smooth variations a_i, p_i and parameters having zero spatial mean. This excludes the entire zero Fourier mode from this problem. Put $D_i^0 = \partial_i + [A_i^0, \cdot\,]$.

(a) Derive the linearized Gauss equation $D_i^0 p_i = 0$ and the gauge variation $a_i \mapsto a_i + D_i^0\varepsilon$. For the Coulomb condition $\partial_i a_i = 0$, compute the eigenvalues of $M = -\partial_i D_i^0$ on each nonzero Fourier mode, using $T_+ = T_1 + \mathrm{i}T_2$, $T_- = T_1 - \mathrm{i}T_2$ and T_3.

(b) For $|\eta| < 1$, prove that M^{-1} maps smooth zero-mean arrays to themselves. Define

$$P = 1 + D^0 M^{-1}\,\mathrm{div}, \qquad P^* = 1 + \nabla M^{-1} D^0 \cdot\,.$$

Prove that Pa is divergence-free and gauge invariant, while $P^* p$ satisfies linearized Gauss. Prove $P^2 = P$. For real smooth zero-mean vector smearings f, g, compute the canonical bracket of $\mathcal{A}[f] = \int\langle f, Pa\rangle$ and $\mathcal{P}[g] = \int\langle g, P^* p\rangle$.

(c) At $\eta = 2$, find a real zero-mean parameter with $M\varepsilon = 0$ but $D^0\varepsilon \neq 0$. Explain why this prevents using the preceding inverse there. Does the calculation establish a global nonlinear reduced bracket?

Solution. At $\Pi^0 = 0$, variation of $D_i\Pi_i$ contains no term $[a_i, \Pi_i^0]$, leaving $D_i^0 p_i$. The linearized residual gauge action is $a \mapsto a + D^0\varepsilon$ and leaves p unchanged. On the mode $e^{\mathrm{i}n\cdot x}T_+$ the first covariant derivative has multiplier $\mathrm{i}(n_1 - \eta)$; the other two have $\mathrm{i}n_2, \mathrm{i}n_3$. Contracting with $-\partial_i$ gives $|n|^2 - \eta n_1$. The T_- and neutral directions give respectively $|n|^2 + \eta n_1$ and $|n|^2$. Reality pairs conjugate modes, so this diagonalization respects real arrays.

For nonzero integer n, $|n_1| \le |n|^2$. Thus all three eigenvalues are at least $(1 - |\eta|)|n|^2 > 0$ when $|\eta| < 1$. Dividing rapidly decreasing Fourier coefficients by them preserves rapid decrease. It therefore defines M^{-1} on smooth zero-mean functions. The real operator M is self-adjoint: the constant D^0 commutes with the spatial derivatives, and their periodic adjoints are their negatives.

Since $\mathrm{div}\, D^0 = -M$, the projection satisfies

$$\mathrm{div}\, P = 0, \qquad PD^0 = 0, \qquad P^2 = P + D^0 M^{-1}\,\mathrm{div}\, P = P.$$

The same commuting constant coefficients give $D^0 \cdot \nabla = -M$. Hence $D^0 \cdot P^* = 0$. Periodic integration by parts identifies the stated P^* as the real pairing adjoint of P. The gradients of the two observables are respectively $P^* f$ in the a variable and Pg in the p variable. Consequently

$$\{\mathcal{A}[f], \mathcal{P}[g]\} = \int \langle P^* f, Pg \rangle = \int \langle f, P^2 g \rangle = \int \langle f, Pg \rangle.$$

The brackets within either family vanish. The Coulomb representative in a gauge orbit is unique: solving $\operatorname{div}(a + D^0 \varepsilon) = 0$ requires $\varepsilon = M^{-1} \operatorname{div} a$ and gives Pa. On constrained momenta $P^* p = p$. The remaining pairing between $\operatorname{im} P$ and $\operatorname{im} P^*$ is nondegenerate. Indeed, for $a = Pa \neq 0$, choosing $p = P^* a$ gives $(a, p) = (Pa, a) = \|a\|_2^2 > 0$; the adjoint argument handles a nonzero p. Thus the bracket just computed is the projected canonical bracket of this linear constraint quotient, rather than an assumption about nonlinear reduction.

Set $\zeta = x^1 + x^2$ and $\varepsilon = T_1 \cos \zeta - T_2 \sin \zeta$. At $\eta = 2$ it gives

$$D_1^0 \varepsilon = T_1 \sin \zeta + T_2 \cos \zeta, \qquad D_2^0 \varepsilon = -T_1 \sin \zeta - T_2 \cos \zeta, \qquad D_3^0 \varepsilon = 0.$$

These derivatives are nonzero, but their ordinary divergence is zero. Thus $M\varepsilon = 0$ and the inverse construction fails. Even in its invertible range, the calculation treats only a linearized problem and excludes zero modes. It proves neither a nonlinear slice nor a global quotient theorem. It supplies an explicit local reduced-bracket calculation within those limits.